SPECTRE: An Event-Driven Vision-Language Tactical Coach for Apex Legends

AI Systems Architecture — Project Proposal

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1 Motivation

Competitive multiplayer games require players to make consequential decisions under uncertainty and severe time constraints. In Apex Legends, an engagement may be decided within seconds by positioning, weapon selection, resource availability, teammate coordination, or the decision to disengage. Players can review recorded matches, but identifying the precise decisions that caused an unfavorable outcome requires substantial experience and careful examination.

Existing approaches to gameplay analysis often emphasize statistics such as kills, damage, accuracy, and match placement. Although these measurements describe performance, they provide limited information about why a particular decision succeeded or failed. A player may lose an engagement despite accurate shooting because of poor cover selection, delayed repositioning, or inadequate coordination with teammates. Conversely, a risky decision may be justified by a temporary numerical advantage or an opportunity to eliminate an opposing squad.

Recent vision-language models (VLMs) provide an opportunity to analyze gameplay visually and generate natural-language feedback. However, applying a general-purpose VLM directly to competitive gameplay presents several challenges. Important information appears in small, specialized HUD elements; visual events develop rapidly; and tactical interpretation depends on game-specific mechanics. A model must therefore understand not only what appears on the screen, but also what that information means in the context of Apex Legends.

This project proposes **SPECTRE**, a system that combines deterministic visual extraction, temporal observations, and vision-language reasoning to produce evidence-based tactical feedback from recorded Apex Legends matches.

2 Project Objectives

SPECTRE aims to transform gameplay recordings into structured observations and actionable tactical assessments. The project pursues four objectives:

1. Extract reliable information from gameplay HUD elements, including player identities, defensive resources, weapons, and screen states.
2. Organize observations chronologically to preserve information about changing gameplay conditions.
3. Combine structured HUD observations with gameplay images to support contextual vision-language reasoning.
4. Generate timestamped assessments explaining significant events, potential decision-making errors, and alternative actions supported by available evidence.

The initial implementation targets **offline analysis**. This choice allows the system architecture and reasoning quality to be evaluated independently of real-time inference constraints. A later extension will investigate event-driven execution and deadline-aware coaching.

3 Proposed System Architecture

The architecture separates perception from tactical interpretation. Rather than requiring a VLM to reconstruct every numerical HUD element, the system first extracts information that can be recognized using conventional computer vision.

The proposed pipeline is:

Gameplay Recording $\longrightarrow$ Frame Sampling $\longrightarrow$ HUD Extraction
$\longrightarrow$ Temporal State Representation $\longrightarrow$ Vision-Language Analysis
$\longrightarrow$ Timestamped Tactical Feedback

This separation is intentional: conventional computer vision handles stable interface elements, while the VLM concentrates on spatial relationships, engagement development, and tactical decision-making.

4 System Components

4.1 Video Ingestion and Sampling

SPECTRE receives an Apex Legends gameplay recording and samples frames at configurable intervals. The initial system uses sparse sampling to establish a functional pipeline. Future experiments will investigate denser sampling around fights, where rapid changes in positioning and combat outcomes require greater temporal resolution.

4.2 HUD Perception

Apex Legends presents structured information through fixed or approximately fixed interface regions. SPECTRE will exploit this property through region-of-interest extraction, visual template matching, color analysis, and selective optical character recognition.

The intended perception targets include squad composition, health and shields, equipped weapons, and interface states such as inventory, map, and spectating. Screen-state detection is particularly important: applying gameplay-specific extractors to unrelated interfaces can produce false observations.

Each detector will therefore operate only when its corresponding screen state is recognized sufficiently confidently.

4.3 Temporal Representation

Individual screenshots cannot establish how an engagement develops. SPECTRE will maintain timestamped observations describing the available information at each sampled frame.

The temporal representation will distinguish between newly observed information and previously observed information. For example, an equipped weapon may remain unchanged across several frames, while its ammunition or the player's shield strength changes continuously.

This representation is intended to help identify transitions, including damage, weapon changes, movement between interfaces, and the emergence of combat events.

4.4 Vision-Language Tactical Analysis

The VLM will receive chronologically ordered images together with corresponding structured observations. Its task is to identify meaningful gameplay events and evaluate decisions using evidence available in the supplied sequence.

The model will be instructed to distinguish direct observations from uncertain interpretations. In particular, it should avoid claiming that an enemy was eliminated, an ability was used, or a tactical mistake occurred when the supplied frames do not support that conclusion.

The intended output is a structured report containing a segment summary, timestamped events, potential mistakes, actionable improvements, and unresolved uncertainties.

5 Evaluation Plan

The prototype will be evaluated along three dimensions.

Perception reliability: Compare extracted HUD information against manually inspected gameplay frames. The analysis will consider correct recognition, missed detections, and false classifications.

Temporal consistency: Examine whether information remains associated with the correct player, weapon, and timestamp as gameplay conditions change.

Coaching usefulness: Evaluate whether the VLM identifies significant events and produces specific, evidence-supported feedback rather than generic advice.

A comparison between VLM-only analysis and HUD-guided VLM analysis is planned to investigate whether structured observations reduce interpretation errors.

6 Expected Contributions and Scope

The primary contribution is an integrated architecture for combining inexpensive, specialized visual extraction with higher-level multimodal reasoning.

SPECTRE is not intended to demonstrate that a general-purpose VLM can already coach competitive players reliably. Instead, the project investigates what additional structure is necessary for such coaching to become useful.

The initial milestone is a working offline prototype with reproducible inputs, intermediate observations, execution logs, and generated reports.

Future extensions may incorporate game-specific knowledge, fine-tuning, fight-aware sampling, and a deadline-aware inference policy that suppresses feedback when it can no longer influence the player's decision.